6.

Ex 6.5. (离散随机变数)

$X \sim Bin(n, p)$ $\quad f_X(x) = \binom{n}{x} p^x (1-p)^{n-x}$

$g(x) = n - x$

$Y = N - X \Rightarrow X = N - Y$

$$f_Y(y) = \sum_{x \in g^{-1}(x)} f_X(x)$$

$$= f_X(n-y)$$

$$= \binom{n}{n-y} p^{n-y} (1-p)^y$$

$$= \binom{n}{y} (1-p)^y p^{n-y}$$

$$= Bin(y, 1-p)$$

$\Rightarrow Y \sim Bin(y, 1-p)$

Ex 6.6

$X \sim Unif(0,1)$ $\quad f_X(x) = \begin{cases} 1 & x \in (0,1) \\ 0 & \text{otherwise} \end{cases}$

$F_X(x) = \int_0^x dx = x$

$Y = g(X) = -\log X \rightarrow$ CDF?

① $\frac{d}{dx} g(x) = -\frac{1}{x} < 0 \quad \forall x \in (0,1)$

$g(x) \searrow$

② $\mathcal{X} = (0,1)$

$\mathcal{Y} = (0, \infty) \quad \forall y > 0, \quad y = -\log x \Rightarrow x = e^{-y}$

$\Rightarrow g^{-1}(y) = e^{-y} \quad \forall y > 0$

③ Thm 6.3 #2.

$F_Y(y) = 1 - F_X(g^{-1}(y)) =$

$= 1 - F_X(e^{-y}) = 1 - e^{-y} \quad \forall y > 0$

$F_Y(y) = 0$. otherwise

Ex 6.7

$X \sim \text{Gamma}(n, \beta)$ $\quad f_X(x) = \begin{cases} \frac{1}{(n-1)!\,\beta^n} x^{n-1} e^{-\frac{x}{\beta}} & x>0 \\ 0 & \text{otherwise} \end{cases}$

$(n \in \mathbb{Z}^+, \beta > 0)$

$Y = g(X) = \frac{1}{X} \rightarrow$ PDF? (Thm 6.4)

① $\mathcal{X} = (0, \infty)$ $\quad y = g(x) = \frac{1}{x}$

$\mathcal{Y} = (0, \infty)$ $\quad x = \frac{1}{y} = g^{-1}(y)$

② $\left| \frac{d}{dy} g^{-1}(y) \right| = \left| -\frac{1}{y^2} \right| = \frac{1}{y^2} \quad \forall\, y>0$

③ $f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$

$= \frac{1}{(n-1)!\,\beta^n} \left(\frac{1}{y}\right)^{n-1} e^{-\frac{1}{\beta y}} \cdot \frac{1}{y^2}$

$= \frac{1}{(n-1)!\,\beta^n} \left(\frac{1}{y}\right)^{n+1} e^{-\frac{1}{\beta y}} \quad \forall y>0$

$f_Y(y) = 0$. otherwise.

Ex 6.8 $\quad X \sim f_X(x)$

$Y = g(X) = X^2 \rightarrow$ CDF? PDF?

X^2 is not monotonic function.

According to Def. of CDF.

$F_Y(y) = P(Y \le y)$

$= P(X^2 \le y) = P(-\sqrt{y} \le X \le \sqrt{y})$

$= P(X \le \sqrt{y}) - P(X \le -\sqrt{y})$

$= F_X(\sqrt{y}) - F_X(-\sqrt{y})$

$f_Y(y) = \frac{d}{dy} F_Y(y)$

$= -\frac{1}{2} y^{-\frac{1}{2}} f_X(\sqrt{y}) - \left(\frac{1}{2} y^{-\frac{1}{2}} f_X(-\sqrt{y})\right)$

$= -\frac{1}{2\sqrt{y}} f_X(\sqrt{y}) - \frac{1}{2\sqrt{y}} f_X(-\sqrt{y})$

$= -\frac{1}{2\sqrt{y}} \left[\underbrace{f_X(\sqrt{y})}_{x>0} + \underbrace{f_X(-\sqrt{y})}_{x<0} \right] \quad \forall\, y>0$

Ex 6.9.

$X \sim N(0,1)$ $\quad f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \quad x \in \mathbb{R}$

$Y = g(X) = X^2$

$\forall\, x<0 \quad g(x) \searrow \quad \rightarrow$

$\forall\, x>0 \quad g(x) \nearrow \quad \rightarrow \quad y \in (0, +\infty)$

① $A_0 = \{0\}$

$A_1 = (-\infty, 0) \quad g_1(x) = x^2 \quad g_1^{-1}(y) = -\sqrt{y} \quad \forall\, y>0$

$A_2 = (0, +\infty) \quad g_1(x) = x^2 \quad g_1^{-1}(y) = \sqrt{y} \quad \forall\, y>0$

② For A_1. $\quad g^{-1}(y) = x = -\sqrt{y}$

$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right| = \frac{1}{\sqrt{2\pi}} e^{-(-\sqrt{y})^2/2} \left| -\frac{1}{2\sqrt{y}} \right|$ Ⓐ

For A_2. $\quad g^{-1}(y) = x = \sqrt{y}$

$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right| = \frac{1}{\sqrt{2\pi}} e^{-(\sqrt{y})^2/2} \left| \frac{1}{2\sqrt{y}} \right|$ Ⓑ

③ $f_Y(y) =$ Ⓐ + Ⓑ $\quad \forall\, y>0$

$= \frac{1}{\sqrt{2\pi}} \frac{1}{\sqrt{y}} e^{-\frac{y}{2}} = \chi^2(1)$ 自由度.

Thm 6.11.

$Y = F_X(X) \quad \forall\, y \in (0,1) \rightarrow$ CDF.

Def of CDF.

$F_Y(y) = P(Y \le y)$

$= P(F_X(X) \le y)$

$= P(F_X^{-1}(\square) \le F_X^{-1}(\square)) \quad F_X^{-1}(\cdot) \nearrow$

$= P(X \le F_X^{-1}(y))$

CDF of X $= F_X(F_X^{-1}(y))$

$= y \quad \forall\, 0<y<1$

$= \text{unif}(0,1)$

$P(Y \ge 1) = 1 \quad \rightarrow Y \sim \text{unif}(0,1)$

$P(Y \le 0) = 0$